Week 11: November 9-November 13, 2026

الأسبوع #: من - إلى

Subject/الموضوع: Science **Unit Title عنوان الوحدة:** Energy Changes in Chemical Reactions **Grade الصف: 7**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1 Exploring the Science Behind Ice Cream	I can explain how thermal energy transfer and freezing point depression help a salt-ice mixture freeze ice cream.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	Inspire Science Unit 1 Workbook (provided by school) and kept in classroom until the end of the unit. Inspire Science Digital Access (not yet available) Teams Class Materials W5 Folder
Lesson 2 Designing and Conducting the Ice Cream Experiment	I can design and conduct a scientific investigation to test how effectively a salt-ice cooling system freezes an ice cream mixture.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.1 Activate prior knowledge using organizers, links, and pre-teaching. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	PPT Presentations, Reading Essentials and Notebook
Lesson 3 Analyzing Results and Explaining Particle Motion	I can organize and interpret experimental data to explain how thermal energy transfer affects particle motion, temperature, and changes of state.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 8.2 Vary challenge level with differentiated tools, resources, and difficulty. 3.1 Activate prior knowledge using organizers, links, and pre-teaching.	PPT Presentations, Reading Essentials and Notebook
Lesson 4 Connecting Science to UAE Sustainability and Finalizing the GRASP Display Board	I can explain how scientific knowledge of thermal energy transfer supports sustainable cooling solutions and communicate my findings through a scientific display board.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 8.2 Vary challenge level with differentiated tools, resources, and difficulty. 3.1 Activate prior knowledge using organizers, links, and pre-teaching.	PPT Presentations, Reading Essentials and Notebook

Homework **الواجبات**	Review lesson notes
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	Please encourage your child to review and practice Science regularly each week using the resources available in their class **Microsoft Teams - class material folder which is organized by week**. The presentations, readings, videos, practice activities, etc. posted in Teams are the primary learning materials and should be used throughout the unit to support understanding and achievement. Science study guides will provide **basic *assessment information, topics, and expectations***; however, they are intended as a guide only and will **not contain detailed study materials,** these are found in the class teams. **Effective preparation for graded formatives and summatives requires students to actively engage with the learning materials provided in their class teams.** Thank you for partnering with us to support your child's learning and success.

Week # الأسبوع#

Technology Integration دمج التكنولوجيا	Inspire Science Digital Interactives, PPT
Cross-curricular links الروابط بين المواد الدراسية	**ELA** – Scientific vocabulary and evidence-based explanations using models and observations.
External Assessment Integration دمج التقييمات الخارجية	MAP
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Thinking – Critical Thinking:** Analyse and evaluate scientific evidence to explain how thermal energy transfer, freezing point depression, and particle motion affect changes in temperature and state, and assess the effectiveness of cooling systems. **Research – Information Literacy:** Gather and interpret scientific information and experimental data to explain endothermic processes, evaluate cooling system designs, and investigate sustainable cooling applications in the UAE.
Standards المعايير	**1. MS-PS1-4:** Develop a model that predicts and describes changes in particle motion, temperature, and state of a pure substance when thermal energy is added or removed. **2. MS-PS1-6:** Undertake a design project to construct, test, and modify a device that either releases or absorbs thermal energy by chemical processes.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can understanding thermal energy transfer and endothermic processes help us design effective cooling systems and promote sustainability?
Success Criteria معايير النجاح **(Measurable)**	**Lesson 1:** **Must – DOK 1:** I can identify an endothermic process and describe the role of salt and ice in cooling. **Should – DOK 2:** I can explain how adding salt lowers the freezing point of water and allows the cooling mixture to become colder. **Could – DOK 3:** I can use scientific reasoning to explain how thermal energy transfers from the ice cream mixture to the salt-ice cooling system.

	Lesson 2: **Must – DOK 1:** I can identify the research question, hypothesis, materials, and safety precautions. **Should – DOK 2:** I can identify independent, dependent, and controlled variables and describe a clear experimental method. **Could – DOK 3:** I can conduct a fair investigation, record accurate measurements, and explain how my experimental design supports reliable results. **Lesson 3:** **Must – DOK 1:** I can organize experimental observations and temperature measurements into a data table and graph. **Should – DOK 2:** I can describe temperature changes and explain how removing thermal energy affects particle motion. **Could – DOK 3:** I can analyze experimental evidence, evaluate the effectiveness of my cooling system, and suggest improvements supported by scientific reasoning. **Lesson 4:** **Must – DOK 1:** I can identify how cooling technologies are used in everyday life and in the UAE. **Should – DOK 2:** I can explain how thermal energy transfer relates to sustainable cooling, traditional Emirati cooling methods, or UAE Net Zero 2050. **Could – DOK 3:** I can evaluate the benefits and limitations of cooling technologies, justify their importance to sustainability in the UAE, and communicate evidence-based scientific conclusions.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Belonging الإنتماء Global Understanding التفاهم العالمي Heritage التراث Domain – المجال: Citizenship (المواطنة) Subdomain – المجال الفرعي: Scientific Innovation and Sustainability (الابتكار العلمي والاستدامة) Area – المجال: Understanding how thermal energy transfer and cooling technologies contribute to scientific innovation, energy efficiency, and sustainable development in the UAE.
	Implementation: Students appreciate the UAE's commitment to scientific innovation and sustainable development by investigating thermal energy transfer through the ice cream experiment. They apply scientific knowledge to design effective cooling systems, analyse experimental evidence, and explore how sustainable cooling technologies support energy efficiency, environmental responsibility, and the UAE Net Zero 2050 initiative.